Reconceptualizing Shadow IT: Motivations, Consequences, and Governance in the Age of Shadow AI

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Zusammenfassung

Eine Kurzzusammenfassung der Vorgehensweise und der wesentlichen Ergebnisse.
Allgemeine Merkmale

- Objektivität: Es soll sich jeder persönlichen Wertung enthalten.
- Kürze: Es soll so kurz wie möglich sein.
- Verständlichkeit: Es weist eine klare, nachvollziehbare Sprache und Struktur auf.
- Vollständigkeit: Alle wesentlichen Sachverhalte sollen enthalten sein.
- Genauigkeit: Es soll genau die Inhalte und die Meinung der Originalarbeit wiedergeben.

Abstract

Obige Zusammenfassung in englischer Sprache.

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List of Abbreviations

AI Artificial Intelligence

BYOD Bring Your Own Device

IT Information Technology

SAI Shadow AI

SIT Shadow IT

SLR Structured Literature Review

1 Introduction - 2 pages

Dieser Teil der Arbeit sollte folgende Inhalte haben: • Einführung in die Problemstellung • Motivation und Herleitung des Themas • Aufbau der Arbeit Hinweis: Es hat sich als hilfreich erwiesen, die Einleitung mit der Zusammenfassung bzw. dem Abstract und der Schlussfolgerung zu vergleichen. Damit stellt man sicher, dass diese inhaltlich im Bezug auf Zielsetzung und Motivation übereinstimmen. Der Umfang sollte ca. 5 % der gesamten Arbeit betragen. (= ca. 2 Seiten)

2 Theoretical Foundations - 6-8 pages

In order to establish a comprehensive understanding of Shadow IT (SIT) and its further, Shadow AI (SAI), this section provides an overview of the theoretical foundations and definitions relevant to these phenomena.

2.1 Shadow IT

2.1.1 Concept and Definition

SIT is a multifaceted phenomenon encompassing the use, development, or deployment of Information Technology (IT) artifacts (e.g., software, hardware, or services) outside the formal control, approval, or awareness of the central IT function. For instance, Haag and Eckhardt (2017, p. 1) describe it as "hardware, software, or services built, introduced, and/or used for the job without explicit approval or even knowledge of the organization," whereas Kopper and Westner (2016, p. 1) emphasize autonomous deployment within business departments without IT involvement. Other authors similarly highlight the lack of alignment with IT governance (Käss et al., 2021, p. 4; Rakovic et al., 2020, p. 4).

Beyond these operational perspectives, SIT can be conceptualized as a structural complement to formal IT, consisting of independently developed and used systems, processes, and practices that are not known, supported, or accepted by the official IT department (Rakovic et al., 2020, p. 147). More concise definitions further highlight either the user-driven development outside central IT or the absence of approval and monitoring as defining features (Rakovic et al., 2020, p. 4; Magunduni and Chigona, n.d., p. 2).

Collectively, these definitions establish SIT as a theoretically grounded construct characterized by autonomy, decentralization, and non-sanctioned IT activity, emerging outside formal IT governance.

2.1.2 Scope

SIT encompasses a wide range of tools and solutions, including spreadsheets, databases, cloud services, mobile devices, and other ad-hoc IT solutions (Rakovic et al., 2020, p. 2; Kopper and Westner, 2016, p. 1). These systems are typically small-scale, developed by individuals or small teams, and tailored to immediate operational needs (Rakovic et al., 2020, p. 2). Their emergence reflects both the accessibility of modern IT tools and the flexibility required by employees in dynamic work environments.

2.1.3 Conceptual Distinctions

While SIT shares similarities with related practices such as workarounds, and Bring Your Own Device (BYOD) programs, it is distinct in being unauthorized and unsanctioned. Workarounds represent behavioral adaptations to organizational constraints, whereas SIT manifests these adaptations through technological artifacts. IT consumerization, on the other hand, refers to the broader trend of user-driven technology adoption, providing a context in which SIT can emerge (Rakovic et al., 2020, p. 2; Haag and Eckhardt, 2017, p. 2).

2.1.4 Core Characteristics

SIT is not necessarily malicious or risky in intention. Users often deploy it to enhance productivity, overcome system limitations, or facilitate collaboration. Its defining features are low visibility, lack of IT support, and a tension between organizational control and individual initiative.

2.2 Shadow AI

Definition of Shadow AI

How Shadow AI differs from Shadow IT

Unique Characteristics

Examples

Emergence and Growth

Scale and Prevalence

Sociotechnical Nature and Governance Challenges

Knowledge Leakage and Data Risk

Governance Drift and Policy Gaps

Organizational Context

Attack Surface and Technical Risks

Ethical Paradox

2.3 Conceptual Background on Shadow Behavior

This section should: • briefly explain why shadow behavior exists in general • stay high-level and theory-driven • avoid listing concrete motivations

Example content: • Workarounds in organizations • Tension between control and autonomy • Informal vs. formal IT usage • Knowledge worker behavior

3 Methodology - 2-3 pages

The review follows a structured approach based on the guidelines provided by Okoli (2015, pp. 887–902) for conducting a Structured Literature Review (SLR) in the field of information systems research. The methodology is organized into four phases of two steps each: Planning, Selection, Extraction, and Execution (Okoli, 2015, p. 885).

While the original framework is designed for collaborative research settings involving multiple reviewers, this review was conducted by a single researcher. Therefore, minor adaptions were made to certain steps (e.g., drafting the research protocol, team training) to ensure methodological rigor while maintaining the integrity of the review process. The following sections describe these phases and their implementation in the context of this review.

3.1 Purpose

The first step in conducting a SLR is to define the purpose of the review, which is to identify and analyze existing literature on the phenomenon of SIT and further the development of SAI in organizational contexts. The SLR aims to synthesize insights on the motivations behind SIT and SAI, their organizational consequences, and the governance approaches employed to manage these phenomena.

3.2 Draft Protocol

The second step includes drafting a research question, a research protocol, and training the team. Since this thesis is conducted by a single person, the protocol and team training steps are not required and will therefore not be carried out further. The research questions guiding this review are as follows:

Main Research Question (RQ): How does SAI differ from traditional SIT in organizations with regard to motivational factors, consequences, and countermeasures?

To better address the research question, it will be broken down into three sub-questions and answered throughout the paper:

1. Which motivational factors known from SIT are transferable to SAI, and which additional motivation factors are specific to SAI?
2. Which organizational consequences of SAI differ from those of SIT?

3. Which countermeasures are discussed for SIT and SAI, and how do they differ in design and expected effectiveness?

3.3 Practical Screening

To make sure only relevant and high-quality literature is included in the review, practical screening criteria are defined to filter the large number of initial search results before reviewing them in detail.

For this review, only journal articles, conference proceedings, and book chapters/sections published in English between 2005 and 2026 with a focus on SIT/SAI in organizational or professional settings were included. The research design of the included studies could be empirical (quantitative, qualitative, or mixed methods), theoretical, conceptual, or review-based, as long as they provide insights relevant to the research questions. The participants in the included studies could be professionals, managers, employees in various roles (e.g., analysts, data scientists, IT staff), or end-users as long as they have a close connection to the organizational context of SIT and SAI.

3.4 Literature Search

The literature search was conducted using four search strings across four databases (AISeL, EBSCO, IEEE, and Web of Science), employing advanced search with Boolean operators (`OR` and `AND`) applied to all fields or title/abstract/keywords depending on the database. The search strings were designed to capture literature on both concepts, as well as their intersection, to ensure a broad and comprehensive coverage of relevant studies. The search strings and their purposes are summarized in Table 1. The search process was completed on March 15, 2026, and therefore includes only literature published up to that date.

Table 1: Literature Search Strings

String	**Purpose**	**Query**
S1	Shadow IT Baseline	`"shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system"`
S2	Shadow AI Baseline	`"shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI"`
S3.1	Shadow IT × Shadow AI	`("shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system") AND ("shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI")`
S3.2	Shadow IT × AI (broader)	`"shadow IT" AND "artificial intelligence"`

String S3.2 was included to capture literature not explicitly using the term *Shadow AI* but discussing the intersection of SIT and Artificial Intelligence (AI) technologies.

Additionally, the search was extended by backward and forward searching of the reference list of the included studies to identify any relevant literature that may have been missed in the initial database search.

3.5 Data Extraction

The remaining literature was subjected to a systematic data extraction process, where the literature was coded and sorted based on their key scope in accordance to the research questions. Based on this preliminary coding, relevant information was extracted and collected in a structured format to group similar concepts and themes together. This process was conducted iteratively, with multiple rounds of coding and extraction to ensure that all relevant information was captured and organized effectively for further analysis and synthesis in the subsequent phases of the review.

3.6 Quality Appraisal

While the practical screening criteria help filter out irrelevant studies before even reading the full-text, the review process also involves a quality appraisal to ensure the included studies meet certain standards. Therefore exclusion criteria need to be defined and applied to the extracted studies (Okoli, 2015, p. 884).

This SLR excluded studies with unsufficient theoretical contribution, unclear methodology or findings, or low prerequisites to adequately answer the research questions.

3.7 Study Synthesization

Study synthesization involves aggregating, summarizing, and interpreting the findings from the included studies to answer the research questions (Okoli, 2015, p. 899). The aggregation utilized the preliminary outline of the review, which was based on the research questions, to group similar concepts and themes, such as definitions, motivations, consequences, and countermeasures of SIT and SAI. [1]

3.8 Writing the Review

The final phase of the review process presents the results in detail, so future reviewers can understand and reproduce the process and the findings (Okoli, 2015, p. 901). Section 4 presents the results of the SLR, organized around the key themes of identified lit-

[1] To support the aggregation and summarization of relevant data, generative AI tools such as ChatGPT (OpenAI, GPT-5.3, April 2026 version) and Claude (Antrophic, Sonnet 4.5, April 2026 version) were used. All outputs were critically reviewed and revised by the author who assumes full responsibility for the final content.

erature, motivational factors, organizational consequences, and governance approaches of SIT and SAI to summarize existing literature and knowledge about these concepts.

4 Structured Literature Review Results - 8-10 pages

4.1 Overview of Identified Literature

In total, 831 records from the four databases were identified. Table 2 provides a breakdown of the search results per string and database, illustrating the initial scope of literature available for review.

Table 2: Database Search Results per String

String	AISeL	EBSCO	IEEE	WoS	Total
S1	296	227	39	94	656
S2	6	79	9	5	99
S3.1	4	2	0	0	6
S3.2	45	18	4	3	70
Total	**351**	**326**	**52**	**102**	**831**

Following the removal of duplicates (124 records) and the application of the practical screening criteria (69 records), 638 records remained for content screening. Reviewing titles and abstracts led to the exclusion of 516 records, leaving 122 records for full-text review and quality appraisal. The full-text review resulted in the exclusion of 41 studies, with 19 additionally excluded due to inaccessibility. The forward and backward searching of references added seven more studies, leading to a final set of 69 included studies for this review.

Table 3 provides a detailed overview of the whole SLR process, including the number of records included, excluded, and remaining at each stage of the screening funnel.

Figure 1 illustrates the temporal distribution of the included studies by publication year, differentiating between SIT and SAI. The results show that SIT has been an established research stream since atleast 2012, with continuous presence and fluctuating publication activity over time, peaking in 2019. In contrast, publication on SAI only begin to emerge in 2021 and remain limited in number. A noticeable increase can be observed from 2024 onwards, with a peak in 2025. Despite this recent growth, the overall volume of SAI remains substantially lower compared to SIT. Overall, the figure highlights that while SIT represents a mature and consistently researched phenomenon, SAI is a comparatively developing area of research.

Table 3: Literature Screening Process

Stage	**Included**	**Excluded**	**Remaining**
Records identified	–	–	831
After duplicate removal	–	124	707
Practical screening			
Language (English only)	–	24	683
Publication year (2005–present)	–	30	653
Publication type	–	15	638
Content screening			
Title screening	254	384	254
Abstract screening	122	132	122
Full-text review			
Not accessible	–	19	103
Full-text screening	62	41	62
Forward searching	3	–	65
Backward searching	4	–	69

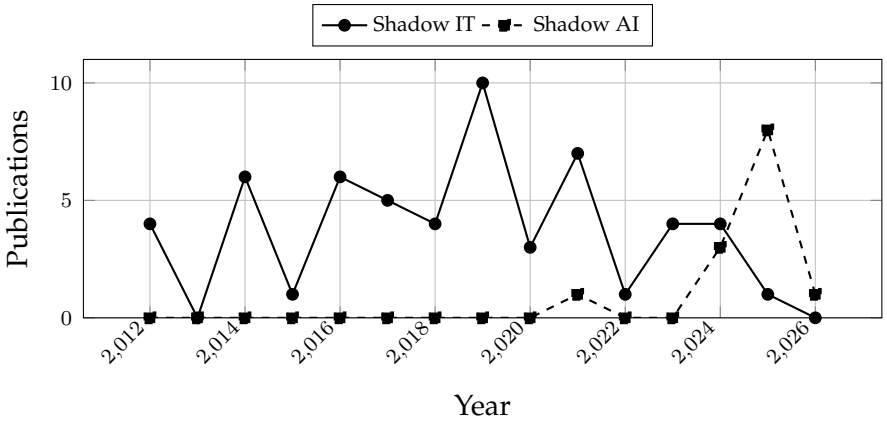

Figure 1: Temporal distribution of included studies on Shadow IT and Shadow AI (2012–2026)

4.2 Motivational Factors

For SAI

4.2.1 Individual Motivations

User Experience and Ease of Use

Percieved Usefulness

Performance Expectancy

Effort Expectancy

4.2.2 Social Motivations

Social Influence

Social Presence

Subjective Norms

4.2.3 Organizational Motivations

IT Department Limitations

System Inadequacies and Business-IT Misalignment

Facilitating Conditions

4.2.4 Technological Motivations

Consumer IT Capabilities and Relative Advantage

4.2.5 Work-Related Stressors

Work Overload

Work Pressure

4.3 Organizational Consequences

4.3.1 Positive Consequences

Innovation Benefits

Individual Performance Improvement

Knowledge Sharing Enhancement

Productivity Gains

Agility and Flexibility

4.3.2 Negative Consequences

Security Risks and Data Breaches

Compliance Violations

Data Leakage

Governance Challenges and Integration Issues

Lack of Awareness

4.4 Governance Approaches and Countermeasures

4.4.1 IT Governance Strategies

Control-Oriented Approach

User-Oriented Approach

Middle Ground/Balanced Approach

4.4.2 Policy Approaches

Communication and Awareness

Understanding and Procedural Justice

4.4.3 Management Strategies

Balancing User Needs and Security

Manager and Leadership Influence

4.4.4 Risk Mitigation & Compliance Mechanisms

Deterrence and Sanctions

Personal and ORganizational Ethics

Cultural Factors

5 Conceptual Development: Extending Shadow IT to Shadow AI - 4-5 pages

REPHRASE

Building on the previously established theoretical foundations and the insights of the SLR results, this section focuses on the conceptual linkage and analytical extension of SIT to SAI. While SIT has been extensively studied over the past decades, resulting in

a well-established understanding of its underlying motivations, organizational consequences, and governance approaches, the emergence of SAI presents a relatively novel phenomenon. Existing literature on SAI is still limited in scope and fragmentation, offering only initial insights withoout a comprehensive conceptual framework.

To address this gap, this section adopts a transfer-oriented perspective. It systematically draws on the established body of knowledge on SIT and examines how its core concepts, theories, and findings can be applied to the context of SAI. This approach is based on the assumption that SAI, while technologically more advanced, represents an evolution rather than a completely independent phenomenon. As such, similar organizational drivers and bevavioral patterns are expected to persist, even though they may manifest differently due to the specific characteristics of AI technologies.

At the same time, the transfer is not assumed to be one-to-one. AI systems introduce new properties, such as increased autonomy, opacity, and scalability, that may alter both the nature and the impact of associated risks and opportunities. Therefore, the analysis not only seeks to identify parallels between SIT and SAI but also to highlight important deviations that emerge in the AI context.

By structuring the conceptual development along the dimensions of motivations, consequences, and governance approaches, this section provides a theoretically grounded foundation for the subsequent analysis. It enables a systematic exploration of SAI despite the current limitations of dedicated literatuer, while ensuring that the insights remain anchored in established research.

REPHRASE

5.1 Transferable Motivations

Performance Improvement/Performance Expectancy

Ease of Use/Effort Expectancy/User-Friendliness

Autonomy Needs/Independence/Self-Determination

Productivity Gains/Efficiency

Social Influence/Peer Influence/Normative Beliefs

Workaround Behavior/Overcoming Constraints

System Limitations/IT Department Limitations

5.2 Novel Motivations

AI Capabilities (NLP, Content Generation)

Accessibility of AI Tools

Generative Capabilities

Advanced Analytical Features

Speed of AI Responses/Rapid Deployment Conversational Interfaces/Natural Language Interaction

AI as Knowledge Assistant/Cognitive Support

Unique Capabilites

5.3 Behavioral Shift in Knowledge Work

Changes in Knowledge Work Patterns/Knowledge Work Transformation

Shift in Employee-Technology Interaction/Human-AI Interaction

AI as Cognitive Assistant/Thinking Partner

Transformation of Decision-Making Processes

Changes in Information Processing/Cognitive Processes

Impact on Professional Expertise/Expert Knowledge

Changes in Work Practices/Work Routines

5.4 Integrated Conceptual Framework

Conceptual models connecting Motivations -> Behaviors -> Outcomes

Theoretical Frameworks for understanding Shadow IT/AI

Relationships between Constructs

Mediating Factors/Mediation Relationships

Moderating Factors/Moderation Effects

Causal Pathways/Causal Chains

Integration of Mulitple Theories

Organizational-Individual Factor Integration

Sociotechnical Integration (Shadow AI)

6 Comparative Analysis of Shadow IT and Shadow AI - 3-4 pages

6.1 Differences in Motivational Drivers

6.2 Differences in Organizational Risk

6.3 Differences in Governance Approaches

7 Discussion - 3-4 pages

8 Limitations

While this review provides a comprehensive overview and valuable conceptual insights into motivations, consequences, and governance approaches of SAI as a emerging topic, it is not without limitations.

First, the literature search was limited to four databases and specific search strings, which may have resulted in the exclusion of other relevant studies that use different terminology or appear in journals not covered by the databases. Although backward and forward searching was conducted to mitigate this risk, it cannot be completely ruled out.

Second, the review was conducted by a single researcher, which may introduce potential bias in the selection, coding, and interpretation of the literature. While efforts were made to ensure rigor and objectivity, the absence of multiple reviewers may limit the reliability of the presented fidings.

Third, the conceptual development and comparative analysis are based on the assumption of a transferability of insights from SIT with the risk of oversimplification in the context of AI due to the different technological characteristics and types of AI tools.

Fourth and finally, the review is limited to literature published to the current date (March 2026), which may not capture all relevant facets of the emerging field of SAI.

9 Future Research Directions

Given the findings and limitations outlined above, there is ample scope for future research on this topic. Empirical studies are needed to validate the transferability of SIT to SAI and to explore the AI-specific characteristics and their impact on organizational processes and outcomes. Qualitative studies, such as case studies or interviews, could provide deeper insights into the changes in organizational practices and employee behavior, while quantitative studies could test the relevance and strength of identified relationships.

Although many forms of SAI can be derived from and build upon the foundations of SIT, AI poses some unique risks that have either already been studied, have not yet been identified, or have yet to emerge. For this reason, more in-depth and specific long-term research is needed.

Overall the field of SAI is still in its early stages, calling for a wide range of research areas (e.g. socio-technical, cybersecurity, ethics) to be explored.

10 Conclusion - 2 pages

In der Schlussfolgerung sollen

• die Themenstellung • der gewählte Ansatz • die Ergebnisse der Arbeit • eine kritische Stellungnahme/Einschätzung • nächste Schritte

deutlich werden. Hinweis: Die Schlussfolgerung sollte mit der Zusammenfassung bzw. dem Abstract und der Einleitung abgeglichen werden. Es sollte immer eine Zusammenfassung der wesentlichen Erkenntnisse der eigenen Arbeit sein, die den Forschungsbeitrag darstellt. Der Umfang der Schlussfolgerung sollte ähnlich wie die Einleitung ca. 5 % der gesamten Arbeit betragen. (= ca. 2 Seiten)

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A Appendix A